

Design and Development of a Risk-Based and Performance-Oriented Medical Equipment Asset Optimization System for Hospital-Level Implementation in Ethiopia

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ABSTRACT

Medical equipment availability and functionality remain persistent challenges in Ethiopian hospitals, where service readiness is constrained by spare part shortages, limited maintenance capacity, incomplete documentation systems, and inconsistent implementation of structured management practices. Although national guidelines emphasize lifecycle-based equipment management, preventive maintenance scheduling, and systematic inventory control, many hospitals lack analytical tools that transform maintenance records into actionable decision-support outputs, resulting in reactive rather than risk-informed maintenance and replacement decisions. This thesis proposes the design and development of a hospital-level Risk-Based and Performance-Oriented Medical Equipment Asset Optimization System for implementation in an Ethiopian hospital. The system integrates established engineering and healthcare management frameworks into a practical software prototype tailored to local operational constraints. Its analytical foundation includes Failure Mode and Effects Analysis (FMEA)-based Risk Priority Number (RPN) scoring for equipment risk prioritization; reliability and maintainability indicators such as Mean Time To Repair (MTTR), Mean Time Between Failures (MTBF), and availability; preventive maintenance compliance measurement; composite index construction using normalization and weighted aggregation; lifecycle cost framing; and multi-criteria replacement prioritization modeling. The study adopts a quantitative design-and-development methodology in which equipment inventory data, corrective maintenance logs, and preventive maintenance records from one hospital will be digitized and structured within a relational database. The system will compute risk, performance, compliance, and replacement indices and generate ranked decision-support outputs, which will be evaluated through computational verification of implemented formulas and sensitivity analysis of weighting schemes. By converting equipment records into structured optimization outputs, the proposed system aims to enhance maintenance prioritization transparency, improve resource allocation, and support proactive lifecycle management within Ethiopian hospital settings.

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List of Abbreviations

A	Availability
AHP	Analytical Hierarchy Process
CI	Composite Indicator
D	Detectability
ERI	Equipment Risk Index
EPS	Equipment Performance Score
FMEA	Failure Mode and Effects Analysis
HTML	HyperText Markup Language
IDE	Integrated Development Environment
ICU	Intensive Care Unit
MTBF	Mean Time Between Failures
MTTR	Mean Time To Repair
$N_{failures}$	Number of Failures
$N_{repairs}$	Number of Repairs
N_{PM}	Number of Preventive Maintenance tasks
O	Occurrence
PMC	Preventive Maintenance Compliance
PM	Preventive Maintenance
RPI	Replacement Priority Index
RPN	Risk Priority Number
RBM	Risk-Based Maintenance
S	Severity
VS Code	Visual Studio Code
$T_{maintenance}$	Total Maintenance Time
$T_{operational}$	Total Operational Time
UI	User Interface
w_i, w_j	Weights in weighted scoring model

CHAPTER ONE

1.1 Introduction

Medical equipment plays a fundamental role in modern healthcare delivery. Critical services such as surgery, intensive care, emergency medicine, diagnostic imaging, and laboratory testing depend on the consistent availability and safe functioning of devices including patient monitors, anesthetic machines, infusion pumps, sterilization systems, and diagnostic analyzers. When equipment becomes unavailable or malfunctions, the consequences may include delayed diagnosis and treatment, interruption of clinical services, compromised patient safety, and inefficient use of limited hospital resources. These impacts are particularly severe in low-resource healthcare environments where redundancy is limited and access to spare parts and technical support may be constrained.

In Ethiopian hospitals, empirical assessments continue to report significant challenges related to medical equipment functionality and availability. Studies conducted in hospitals within the Amhara region indicate notable proportions of non-functional or improperly installed equipment, with contributing factors including spare-part shortages, limited maintenance workshop capacity, inadequate documentation, and insufficient training of users and biomedical staff [1], [2]. Additional facility-level research shows that procurement practices, storage conditions, and broader equipment management structures also influence equipment performance, indicating that functionality problems often arise from weaknesses across multiple stages of the equipment lifecycle rather than isolated maintenance failures [3].

National guidance, including the Ethiopian Hospital Services Transformation Guidelines, emphasizes structured inventory systems, preventive maintenance implementation, equipment history documentation, and formalized processes for corrective maintenance and equipment disposal [4]. While such policies establish clear expectations for equipment management, evidence suggests that systematic and data-driven management practices remain inconsistently implemented in many healthcare facilities [3]. Under resource constraints, biomedical teams frequently prioritize maintenance activities reactively, without analytical tools that help quantify equipment risk, performance burden, or replacement urgency.

International healthcare technology management literature further highlights the importance of lifecycle-based equipment management supported by performance monitoring and data-driven maintenance strategies [5]. Reliability assessment methods and maintenance analytics demonstrate that structured equipment records can be transformed into measurable indicators that support maintenance optimization and improved service continuity [6], [7]. These perspectives emphasize the potential for transitioning from documentation-based equipment management toward decision-support systems that use quantifiable performance and risk indicators to guide hospital equipment management.

1.2 Problem Statement

Recent evidence indicates that Ethiopian hospitals continue to experience significant limitations in medical equipment availability and functionality due to systemic management and resource constraints[1]. An assessment across public hospitals found substantial rates of non-functional equipment and identified documentation quality, training, and institutional capacity as important determinants of equipment functionality [2]. Additional facility-level studies further show that procurement practices, storage conditions, and utilization processes influence equipment performance, indicating that equipment outcomes reflect weaknesses across multiple stages of lifecycle management rather than isolated technical failures [3].

National policy frameworks emphasize structured equipment inventory management, preventive maintenance scheduling, calibration oversight, and formalized corrective maintenance and disposal processes in hospitals [4]. To support these goals, the Ethiopian Ministry of Health introduced the Medical Equipment Management Information System (MEMIS), which enables hospitals to register equipment inventories, record maintenance activities, and generate reports across national, regional, and facility levels [17]. Similarly, recent work conducted at Addis Ababa University proposed a web-based platform to improve hospital-level equipment documentation and maintenance tracking [18]. While these systems improve equipment information visibility and record management, they primarily focus on documentation and reporting and do not incorporate analytical decision-support capabilities for evaluating equipment performance or prioritizing maintenance and replacement decisions. At the same time, the healthcare technology management literature provides analytical frameworks such as reliability metrics, risk-based maintenance

approaches, and multi-criteria decision models that can support evidence-based equipment management [7], [9], [10], [14].

Despite the availability of these analytical methods, they are rarely integrated into practical digital tools used by hospitals for routine equipment management. As a result, biomedical engineering teams often operate under resource constraints without structured tools that quantify equipment risk, performance burden, preventive maintenance compliance, or replacement urgency.

This thesis therefore addresses the implementation gap by developing a hospital-level Medical Equipment Asset Optimization System that integrates risk scoring, reliability metrics, preventive maintenance monitoring, and multi-criteria replacement prioritization into a structured prototype demonstrated using real hospital inventory and maintenance data.

1.3 Objectives

1.3.1 General Objective

To design and develop a **Risk-Based and Performance-Oriented Medical Equipment Asset Optimization System for hospital-level implementation in Ethiopian healthcare facilities**, integrating risk prioritization, reliability performance metrics, preventive maintenance compliance monitoring, and replacement prioritization models to support proactive and data-driven equipment management.

1.3.2 Specific Objectives

- To design a structured framework for integrating medical equipment inventory, maintenance history, and operational data required for analytical equipment management.
- To develop analytical models that quantify equipment performance, reliability, and maintenance effectiveness using indicators such as MTTR, MTBF, availability, and preventive maintenance compliance.
- To implement risk-based and multi-criteria decision models for prioritizing equipment maintenance and replacement using risk prioritization and lifecycle considerations.

- To develop and evaluate a prototype decision-support system that generates equipment performance indicators, risk rankings, and replacement prioritization outputs using hospital equipment data.

1.4 Scope of the Project

This project focuses on the design and prototype development of a hospital-level medical equipment asset optimization system for equipment managed by biomedical engineering departments. The scope includes equipment inventory records, maintenance history, and preventive maintenance activities across major clinical areas such as intensive care, operating theaters, emergency services, imaging, inpatient wards, and laboratories.

The system will analyze these records to generate equipment management indicators including reliability metrics, preventive maintenance compliance, risk-based prioritization, and replacement ranking. The prototype will be demonstrated using equipment and maintenance data from an Ethiopian hospital to evaluate feasibility. The study does not include national system integration, clinical decision support, or patient-identifiable data.

1.5 Significance of the Project

Medical equipment availability is a critical determinant of hospital service readiness. Studies conducted in Ethiopian hospitals continue to report meaningful levels of non-functional equipment and operational barriers that limit effective utilization and maintainability [3]. At the same time, national policy frameworks require hospitals to maintain equipment inventories, implement preventive maintenance programs, and document maintenance activities to support systematic equipment management [4]. However, in many hospitals equipment records are maintained primarily for documentation purposes and are not consistently used to support structured prioritization and planning decisions.

This project contributes to addressing this gap by integrating analytical decision-support capabilities into a hospital-level equipment management prototype. The system introduces risk-based prioritization and performance monitoring using reliability indicators and preventive maintenance compliance measures to help biomedical engineering teams identify equipment that requires urgent attention. In low-resource healthcare environments where maintenance capacity,

spare parts, and technical staff are limited, such prioritization tools are particularly valuable because they help direct available resources toward equipment whose failure would have the greatest operational impact.

The system also supports structured equipment replacement prioritization, allowing hospitals to identify assets that impose high maintenance burdens or operational risk. By providing clear equipment performance indicators and prioritization outputs, the prototype supports more transparent and evidence-based equipment management practices. Overall, the project demonstrates how analytical methods can be translated into a practical decision-support tool for biomedical engineering departments and hospital administrators, contributing to improved equipment management practices in resource-constrained healthcare settings.

CHAPTER TWO

2.1 Literature Review

Effective management of medical equipment requires a lifecycle approach that includes planning, procurement, operation, maintenance, and eventual replacement or disposal. Ethiopian national policy frameworks emphasize this lifecycle perspective and define governance structures and institutional responsibilities for hospital-level medical equipment management [4], [5]. However, empirical evidence from Ethiopian hospitals indicates persistent challenges affecting equipment availability and utilization. Studies report that failures across several stages of the equipment lifecycle contribute significantly to reduced functionality and underutilization of medical devices in public hospitals [1], [2], [3]. For example, research conducted in comprehensive specialized hospitals in the Amhara region identified a considerable proportion of non-functional equipment, with shortages of spare parts identified as major barriers to repair and restoration [1]. Other studies also found that equipment functionality is strongly associated with factors such as the availability of service manuals and adequate user training, highlighting the importance of documentation and technical competence in maintaining equipment performance [2].

To improve coordination and visibility of medical equipment information across the healthcare system, the Ethiopian Ministry of Health introduced the Medical Equipment Management Information System (MEMIS), a web-based platform that enables equipment registration, maintenance documentation, reporting, and procurement requests across national, regional, and facility levels [17]. Similarly, a web-based medical equipment management system developed by biomedical engineering students at Addis Ababa University aimed to improve hospital-level equipment tracking and maintenance documentation through centralized digital records [18]. While these systems improve equipment data management and information accessibility, they primarily focus on documentation and reporting and lack analytical decision-support tools for evaluating equipment performance or prioritizing maintenance and replacement decisions.

Beyond information management systems, the literature proposes several analytical frameworks that can support more evidence-based medical equipment management. Reliability engineering metrics such as availability and maintainability indicators provide quantitative measures for evaluating equipment operational performance and maintenance effectiveness [7], while predictive

maintenance and maintenance data analytics allow hospitals to derive performance indicators from maintenance records to improve maintenance planning and resource allocation [6]. Risk-based maintenance approaches further prioritize maintenance actions by evaluating the likelihood and impact of equipment failure, and structured methods such as Failure Mode and Effects Analysis (FMEA) provide quantitative risk prioritization through severity, occurrence, and detectability scoring [8], [9], [11]. In addition, decision-support approaches such as composite indicators, lifecycle cost analysis, and multi-criteria decision-making models enable multiple performance, risk, and cost factors to be integrated into interpretable scores that support maintenance planning and equipment replacement prioritization [10], [14]. These analytical methods provide a foundation for incorporating decision-support capabilities into equipment management systems, addressing the analytical limitations identified in existing platforms [12], [15], [16].

Although existing platforms such as MEMIS and similar hospital-level systems improve the documentation and visibility of equipment information, they primarily focus on data registration and reporting rather than analytical decision support. At the same time, the literature provides well-established analytical methods for evaluating reliability, risk, maintenance performance, and replacement decisions. However, these analytical approaches are rarely integrated into practical digital tools used by hospitals for everyday equipment management. Therefore, there remains a need for systems that combine equipment management data with analytical models to support evidence-based decision-making in hospital medical equipment management.

CHAPTER THREE

3.1 Methodology

3.1.1 Materials

The development of the prototype system requires a set of software tools for application development, data management, visualization, and documentation. Table 1 summarizes the main components and technologies selected for implementing the system.

Table 1. Software Requirements

Component / Material	Selection Choice for the Project
Development Environment (IDE)	Visual Studio Code (VS Code)
Programming Language	Python / JavaScript
Backend Framework	Flask / Supabase
Frontend Technologies	HTML, CSS, next.js, react.js
Database Management System	MySQL (relational database) / postgres
Visualization Library	Chart.js
Documentation Tools	Microsoft Word

3.1.2 Methods

Study Design

This thesis will use a quantitative, design-and-development approach. The work consists of designing the analytical framework, building a software prototype that implements explicit published formulas, and evaluating the prototype using real equipment-level data from a hospital. The study does not involve qualitative interviews as a formal data source; instead, it relies on structured equipment inventory and maintenance records available in the selected hospital.

The project will be implemented in an Ethiopian hospital with hospital-wide equipment coverage. The selection will depend on access, permission and availability of basic equipment and maintenance records. The hospital implementation will focus on equipment managed by or relevant to the biomedical engineering function, including devices used in ICU, operating rooms, emergency services, imaging, wards, and laboratories.

Figure 1 presents the conceptual workflow of the proposed system, showing how equipment inventory and maintenance data are analyzed to generate performance indicators and prioritization outputs for equipment management.

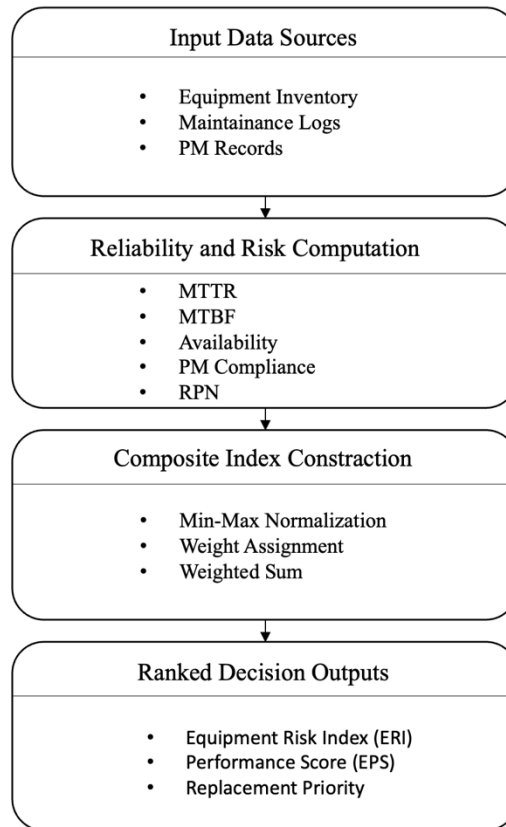


Figure 1. Overall methodological framework

Data Collection

Data collection will be limited to equipment-level operational and maintenance data. The minimum dataset required includes: an equipment inventory list (device type/category, department, identification code, installation year if available), maintenance and failure logs (dates, reported fault, actions taken), repair time measures (start/end time or total repair time per event), and preventive maintenance schedule/completion records. If available, additional fields such as spare part usage, maintenance cost proxies, and service contract status will be captured. The study

will not collect patient data. Data will be digitized into the system database to enable calculation of the analytical metrics described below.

Risk Prioritization Using FMEA Risk Priority Number

Risk-based prioritization in this system will draw on Failure Mode and Effects Analysis (FMEA) logic and risk-based maintenance theory. FMEA calculates a Risk Priority Number (RPN) as the product of three components: occurrence likelihood (O), severity (S), and detectability (D). In an applied healthcare FMEA study, RPN is calculated using the explicit equation:

$$RPN = S \times O \times D \quad (1)$$

with O, S, and D scored on defined scales (commonly 1–10) [11]. In this thesis system, the RPN structure will be adapted to medical equipment risk prioritization. The adaptation is not to change the equation but to define how O, S, and D will be assigned using equipment-level data and biomedical engineering context. Severity (S) will represent clinical criticality of the device and its potential consequence of failure in the department where it is used. Occurrence (O) will represent the frequency or likelihood of failure, estimated from historical maintenance logs as the number of breakdown events observed over a defined period or as a derived annualized failure rate when feasible [6]. Detectability (D) will represent the ability to detect or mitigate failures early; in equipment management, detectability can also incorporate mitigation capacity such as availability of backup equipment or alternative devices, because low detectability and low redundancy both increase operational risk. This mapping aligns with reliability assessment literature that includes device criticality and backup availability within “availability and readiness” considerations, and with risk-based maintenance approaches emphasizing criticality assessment and prioritization of prevalent failure modes [7], [9].

Figure 2 illustrates the structure used to compute Risk Priority Numbers (RPN) for equipment prioritization.

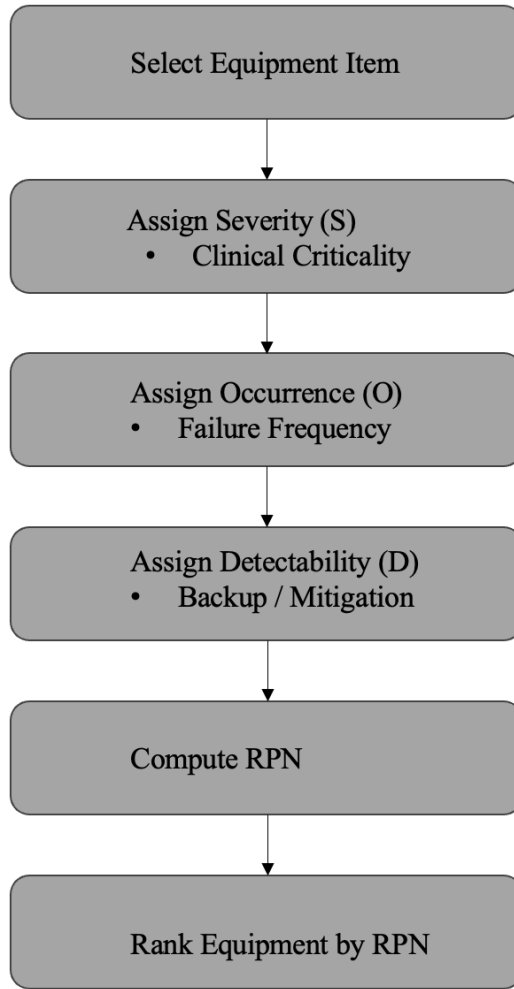


Figure 2. Risk scoring process

Reliability and Availability Metrics

To quantify performance, the system will compute reliability and maintainability metrics using explicit equations for MTTR, MTBF, and availability. A medical equipment maintenance analytics study provides explicit equations for these metrics, including availability as a function of MTBF and MTTR [12]. The equations implemented are:

$$A = \frac{MTBF}{MTBF + MTTR} \quad (2)$$

$$MTBF = \frac{T_{operational}}{N_{failures}} \quad (3)$$

$$MTTR = \frac{T_{maintenance}}{N_{repairs}} \quad (4)$$

Where $T_{operational}$ is the total operational time during the observation period, $N_{failures}$ is the number of failures, $T_{maintenance}$ is the total maintenance time (repair durations), and $N_{repairs}$ is the number of repair events [12]. In the pilot hospital context, operational time may be approximated using observation period length when continuous operational logging is absent, while repair durations can be computed from recorded maintenance tasks or from calculated downtime durations. If device-level operational hours are not available, the system will compute MTBF using available failure intervals (e.g., days between recorded failures) or compute at category level. All approximations will be documented explicitly as part of evaluation transparency.

Performance outputs will include device-level or category-level MTTR values and availability estimates, as well as dashboards summarizing low-availability devices and devices with high repair time burdens. Reliability assessment literature indicates that such performance assessment supports strategic maintenance decisions, increases availability, and optimizes resource allocation [7].

Preventive Maintenance Compliance

Preventive maintenance compliance (PMC) will be computed as a percentage of scheduled preventive maintenance work that is completed within a defined time period. A maintenance management study defines PMC as the percentage of preventative work scheduled and completed in a set time period and provides an explicit example calculation [13]. The equation implemented is:

$$PMC = \frac{N_{PM\ completed}}{N_{PM\ scheduled}} \times 100 \quad (5)$$

where $N_{PM\ completed}$ is the number of preventive maintenance work orders completed and $N_{PM\ scheduled}$ is the number of preventive maintenance work orders scheduled in the period of interest [13]. In this thesis system, PMC will be computed monthly and optionally per department or equipment category, depending on data availability. This aligns with national guidance emphasizing preventive maintenance scheduling, log sheets, and mechanisms for ensuring tasks are performed according to schedule [4]. PMC outputs will be used to highlight compliance gaps and overdue preventive tasks, providing a management indicator for preventive maintenance discipline.

Composite Scoring, Weighting, and Normalization

To produce interpretable indices and rankings that combine multiple indicators, the system will employ weighted scoring and normalization. Composite indicator methodology emphasizes that normalization is required before aggregation when indicators have different units and that different aggregation and weighting methods can substantially affect rankings, motivating robustness and sensitivity analysis [14]. In the system, normalization of indicator values to a comparable scale will be implemented using min–max normalization as explicitly provided in an AHP-based replacement prioritization model, which defines normalization as:

$$NormScore = \frac{ScoreValue - MinimumValue}{MaximumValue - MinimumValue} \quad (6)$$

[16]. This normalization method scales values to a 0–1 range, supporting aggregation of disparate metrics such as MTTR (hours), availability (unitless ratio), or cost proportion.

Weighted scoring will be implemented using the general total score formula defined in a medical equipment replacement prioritization model. The model defines the total score for the i -th device as a weighted sum of criterion scores:

$$TS_i = \sum_{j=1}^m w_j s_{ij} \quad (7)$$

where w_j is the weight of criterion j , s_{ij} is the score of device i on criterion j , and m is the number of criteria [16]. Although [16] applies this model to replacement prioritization, the formula provides a general, explicit method for aggregating multiple scored criteria into a single index. This thesis system will use this structure for aggregated indices when needed, such as a Performance Score that combines normalized availability, MTTR efficiency, downtime burden proxies, and preventive maintenance compliance, subject to available data.

Weight selection will be guided by the practical importance of criteria for clinical service continuity and patient safety, and it will be justified using both domain reasoning and robustness checks. Composite indicator methodology emphasizes that weighting and aggregation choices introduce uncertainty, and it recommends performing uncertainty and sensitivity analyses to evaluate how changes in weights affect ranking outcomes [14].

Therefore, the system evaluation will include sensitivity checks where weights are varied within reasonable ranges and stability of top-ranked priorities is assessed.

Figure 3 presents the process used to normalize indicators and compute overall equipment prioritization rankings.

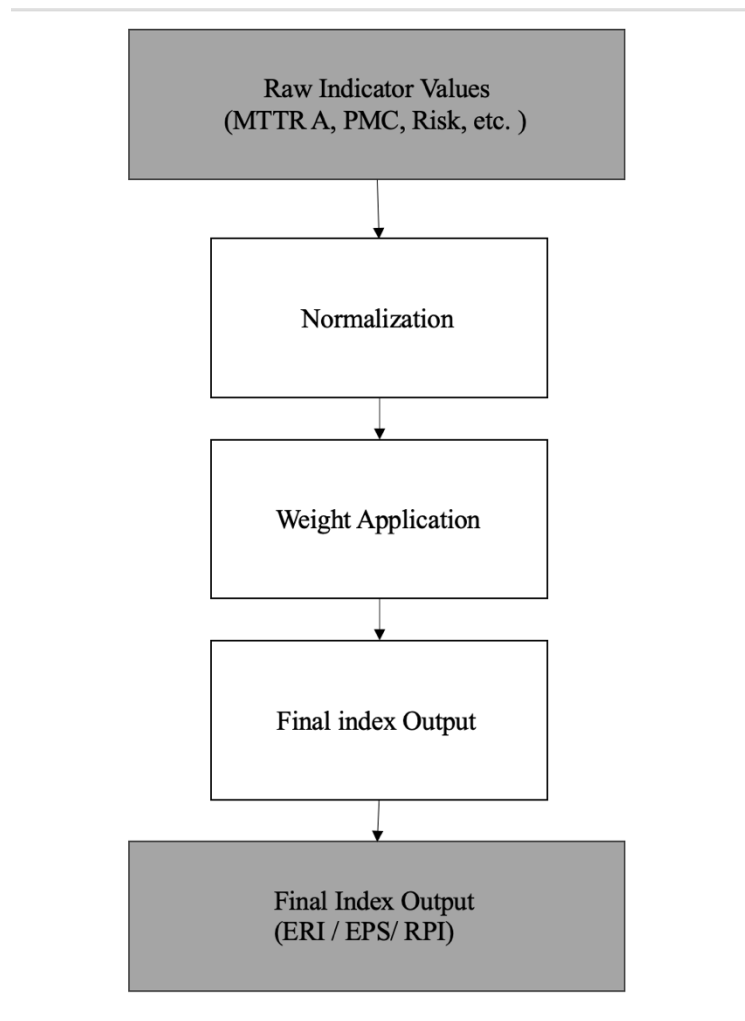


Figure 3. Composite indicator computation process

Replacement Prioritization Model

Replacement prioritization will be implemented as a multi-criteria decision-support output using weighted scoring and normalization. Replacement decision literature argues that hospital managers need quantitative tools to prioritize replacement of equipment and that multi-criteria decision-making approaches address ambiguity in defining and combining indicators by assigning weights and computing overall scores for decision alternatives [10]. The replacement module in this thesis will implement a scoring structure consistent with the weighted sum model in (7) and normalization in (6) [16], while drawing on replacement decision modeling concepts and indicator system design principles described in medical equipment replacement decision literature [10]. The replacement module will incorporate criteria aligned with lifecycle and equipment performance considerations, such as age, failure frequency, availability, maintenance burden, service support or spare part availability proxies when available, and risk priority outputs derived from RPN. The final replacement priority index will be derived from normalized total scores, consistent with the approach described in [16].

Software Architecture

The system will be developed as a web-based prototype suitable for hospital-level deployment. The architecture will be designed around three main layers: a user interface layer, an application logic layer, and a database layer. The user interface layer will enable equipment registration, maintenance event entry, preventive maintenance scheduling and completion entry, and dashboard viewing. The application logic layer will implement the computation engine for the analytical models described previously including equations (1)–(7). The database layer will store inventory records, maintenance event logs, preventive maintenance schedules, computed indicators, and generated rankings.

The system design will incorporate principles of data structuring and auditability emphasized in biomedical maintenance analytics literature. Data-driven maintenance studies highlight that fragmented data sources, non-standard codification, and poor interoperability hinder maintenance decision support and that structured datasets and standardized indicators enable reproducible comparisons and more advanced analytics [6]. Therefore, the system will emphasize consistent data entry fields, standardized equipment category labels, and ability to export reports for administrative review. The system will also be designed to function with partial datasets by

allowing optional fields; this is essential in contexts where maintenance records may not be complete and where standardized indicators may not have been historically tracked [6].

The proposed system follows a layered software architecture to separate data management, application logic, and user interface components. Figure 4 illustrates the layered architecture used to implement the medical equipment asset optimization system.

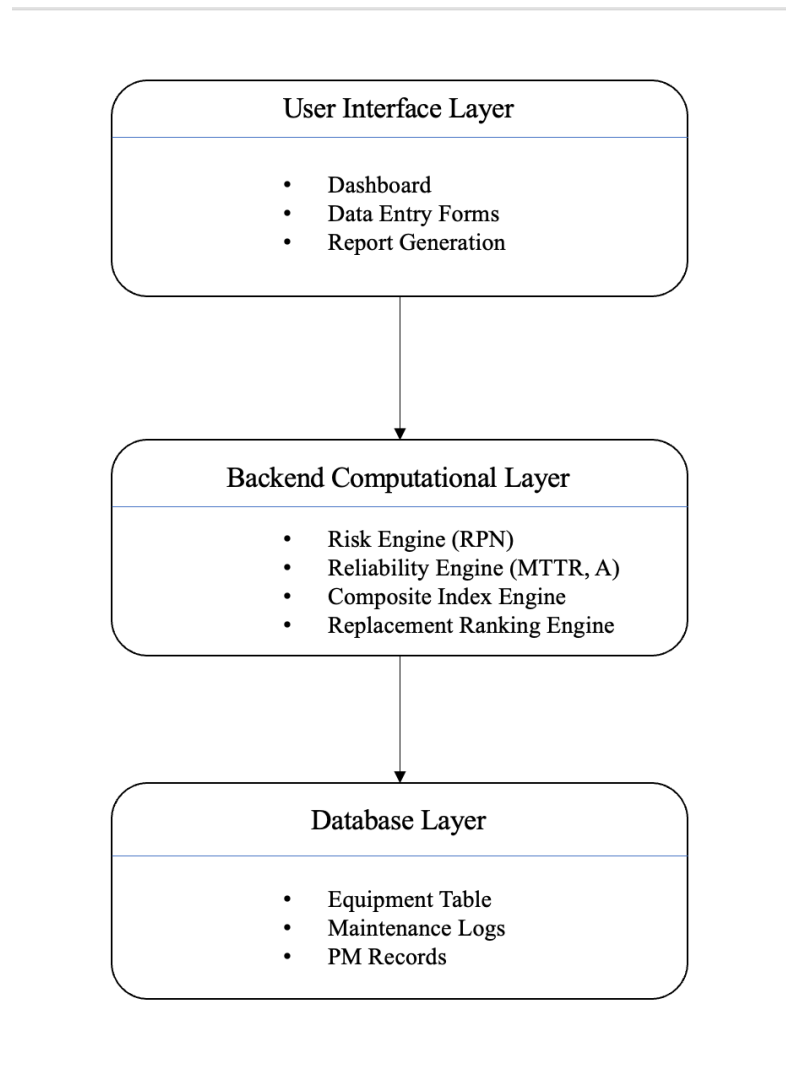


Figure 4. Layered software architecture.

Evaluation Strategy

Evaluation of the system will be quantitative and prototype-focused. Three levels of evaluation will be performed.

First, computational verification will be conducted. A sample of devices will be selected and manual calculations will be performed for RPN (equation 1), reliability and availability metrics (equations 2–4), preventive maintenance compliance (equation 5), normalization (equation 6), and weighted scoring outputs (equation 7). System outputs will be compared against manual calculations to confirm that the system implements published formulas correctly.

Second, robustness and sensitivity checks will be performed for weighting-based indices. Composite indicator methodology emphasizes that weighting and aggregation choices influence rankings, and it recommends sensitivity analysis to evaluate ranking stability under alternative modeling assumptions [14]. Therefore, the project will vary key weights within plausible ranges and assess whether the identity of high-priority devices remains stable. Stability in the top-ranked priorities will support the robustness of the prioritization logic for practical use.

Third, practical demonstration will be conducted using the real world hospital dataset. The system will generate ranked outputs, including a high-risk equipment list (based on RPN outputs), low availability and high-maintenance-burden list (based on MTTR and availability), preventive maintenance compliance reports and overdue task lists, and replacement priority rankings (based on multi-criteria scoring). Outputs will be reviewed for engineering plausibility and for alignment with known high-criticality environments, such as ICU and operating theatres, where equipment failure often has high clinical consequence and where maintenance analytics studies have reported higher failure rates and downtime burdens in critical units [6].

3.2 Work Plan

The implementation of the project will follow the phased timeline shown in Table 2, which outlines the sequence and expected duration of key research and development activities.

Table 2. Project Timeline

№	Activities	Feb		Mar				Apr				May				Jun	
		W3	W4	W1	W2	W3	W4	W1	W2	W3	W4	W1	W2	W3	W4	W1	W2
1	Project Planning																
2	Literature Review																
3	Proposal Writing & Defense																
4	Analytical Model & Database Design																
5	Hospital Permission & Data Coordination																
6	Data Collection																
7	Software Prototype Development																
8	System Integration & Sensitivity Testing																
9	Results Generation																
10	Thesis Preparation																
11	Final Defense																

3.3 Cost Estimation

The estimated financial requirements for completing the project are summarized in Table 3, covering transportation, communication, stationery, and other miscellaneous operational expenses.

Table 3. Budget Breakdown

No.	Budget Item	Estimated Unit Cost (ETB)	Quantity	Total Cost (ETB)
1	Transport	360	5trips	1,800
2	Internet and Communication	—	—	1,000
3	Stationery	—	—	1,000
4	Miscellaneous	—	—	2,500
Total Estimated Budget				6,300 ETB

3.4 Conclusion and Future Work

Conclusion

This proposal presents the design and development of a hospital-level Risk-Based and Performance-Oriented Medical Equipment Asset Optimization System aimed at improving medical equipment management in Ethiopian hospitals. The system integrates established analytical approaches including FMEA-based risk prioritization, reliability and maintainability indicators (MTTR, MTBF, and availability), preventive maintenance compliance monitoring, and multi-criteria scoring methods to generate structured prioritization outputs for equipment maintenance and replacement. By transforming routine equipment inventory and maintenance records into measurable indicators and ranked decision-support outputs, the proposed prototype seeks to support biomedical engineering teams in allocating limited maintenance resources more effectively and improving equipment service readiness in resource-constrained hospital environments.

Future Work

Future work may extend this prototype through validation across multiple hospitals to evaluate the robustness and generalizability of the prioritization framework. Additional development may include integration with existing hospital information systems such as MEMIS, expansion of equipment datasets, and incorporation of advanced analytics such as predictive maintenance models for early failure detection. These extensions would further strengthen the role of data-driven decision support in improving medical equipment management and operational sustainability in low-resource healthcare settings.

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